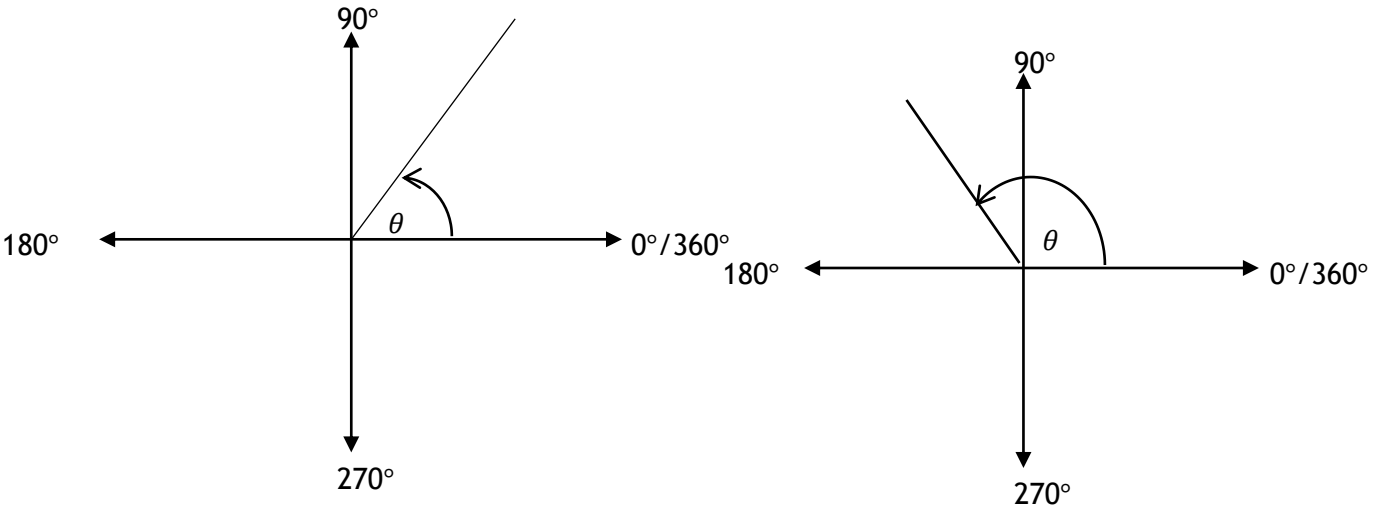
TRIGONOMETRY ON THE CARTESIAN PLANE

We now have to place the right angled triangle on the Cartesian plane. When we move the triangle around in the Cartesian Plane, there are a few things that change.

All angles in the Cartesian Plane are measured from the x –axis in an ***anti-clockwise*** direction



Watch this video showing the ratios in Quadrant 1:

https://www.youtube.com/watch?v=SqrArq-AbR0&t=30s *(4:26 min)*

GEOGEBRA INVESTIGATION:

https://www.geogebra.org/m/CR3JxVh6 (Focus on sin, cos and tan only)

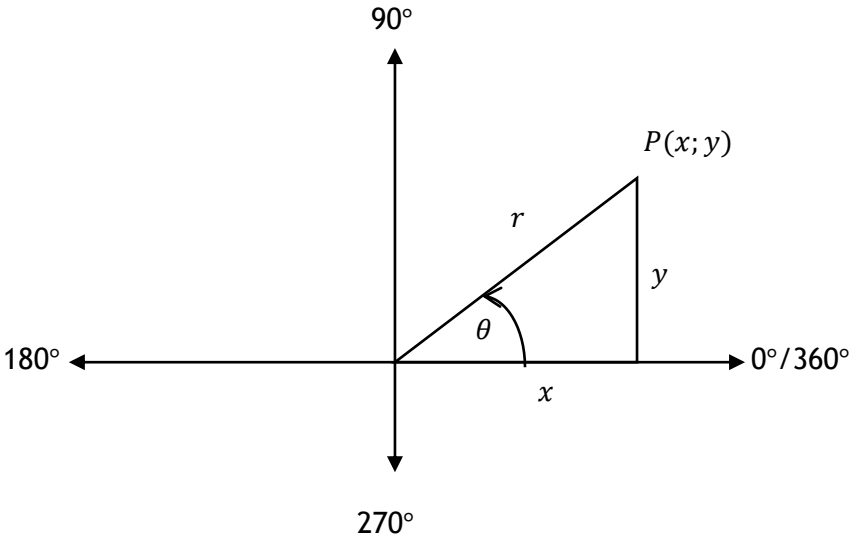
1. Do the **values** of the trigonometric functions change when the radius changes? No
   Explain The angle does not change.

2. Set the **radius to 1** and increase the angle slowly from 0° *to* 360°. Observe how the **values** of the ratios change as the radius rotates in an anti-clockwise direction. Write down any special trigonometric function values that you notice.
   at 0°, 90°, 180°, 270°, 360° some ratios are 0, 1, -1

3. Observe how the **signs** of the trigonometric functions change in different quadrants as the angle increases.
   Now complete the tables for each quadrant:

Quadrant 1

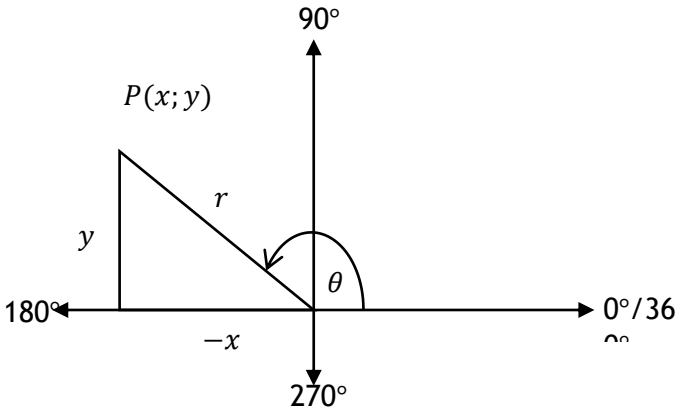
- x is positive
- y is positive
- r is always positive
- $0° < \theta < 90°$



Function	Ratio	" + " or " – "
$\sin\theta$	$\frac{y}{r}$	+
$\cos\theta$	$\frac{x}{r}$	+
$\tan\theta$	$\frac{y}{x}$	+

Quadrant 2

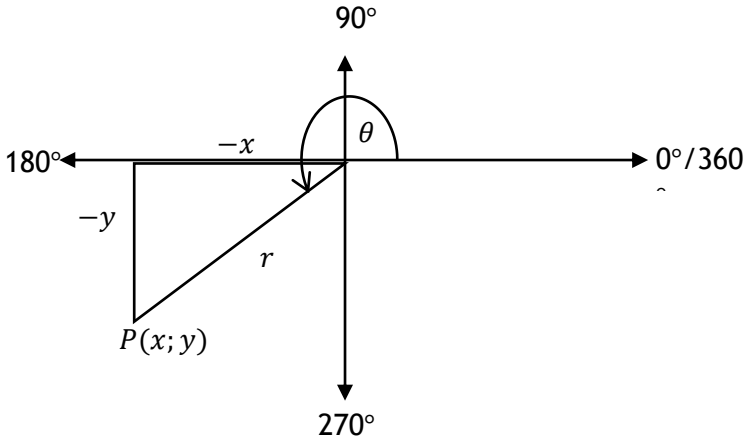
- x is negative
- y is positive
- r is always positive
- $90° < \theta < 180°$



Function	Ratio	" + " or " – "
$\sin\theta$	$\frac{y}{r}$	+
$\cos\theta$	$\frac{-x}{r}$	–
$\tan\theta$	$\frac{y}{-x}$	–

Quadrant 3

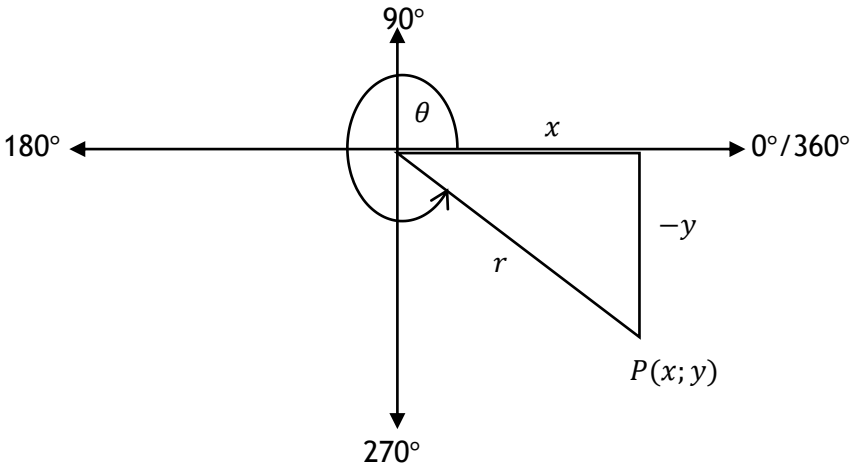
- x is negative
- y is negative
- r is always positive
- $180° < \theta < 270°$



Function	Ratio	" + " or " – "
$\sin\theta$	$\frac{-y}{r}$	–
$\cos\theta$	$\frac{-x}{r}$	–
$\tan\theta$	$\frac{-y}{-x}$	+

Quadrant 4

- x is positive
- y is negative
- r is always positive
- $270° < \theta < 360°$



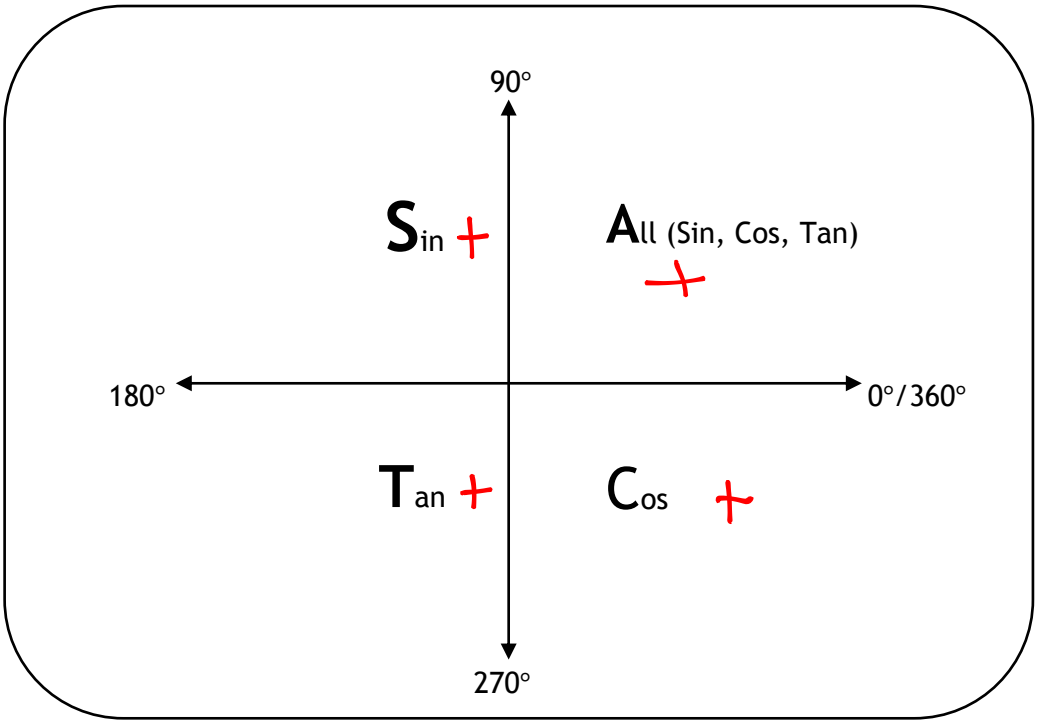
Function	Ratio	" + " or " – "
$\sin\theta$	$\frac{-y}{r}$	–
$\cos\theta$	$\frac{x}{r}$	+
$\tan\theta$	$\frac{-y}{x}$	–

CAST Diagram

The CAST diagram summarises the information regarding the 3 trigonometric functions in each of the quadrants in the Cartesian plane.

We look to see which functions are positive in each quadrant.

Reading from quadrant 4 anti-clockwise we get the "**CAST**" diagram.



ACTIVITY:

Click on the links below and follow the instructions:

https://www.geogebra.org/m/hbPBPDBF Using Opposite, Adjacent and Hypotenuse in the CAST diagram.

SKETCH PROBLEMS (Pythagoras type Problems)

*(These problems are done without the use of a calculator. - Always use **CAST** rule)*

Example 1:

Given that $\cos\theta = \frac{1}{2}$ find **without the aid of a calculator** the values of:

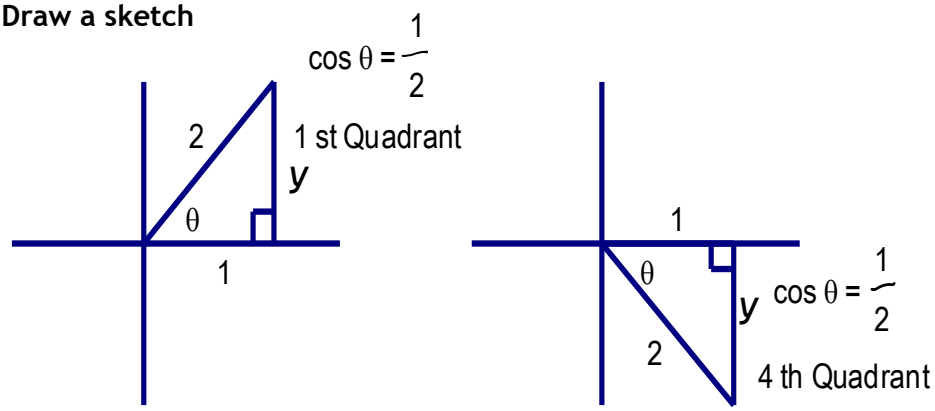
(i) sin Θ

(ii) tan² Θ if:

Step 1: Decide which quadrant/s to work in. Sometimes it is only one quadrant.

Remember: You have TWO quadrants where cos Θ >0 and $\cos\theta = \frac{a}{h} = \frac{x}{r}$ (Quadrants 1 and 4)

Step 2: Draw a sketch



Note: the hypotenuse is ALWAYS positive

Step 3: Use the theorem of Pythagoras to find y:

$$x^2 + y^2 = r^2$$
$$1^2 + y^2 = 2^2$$
$$\therefore\ y = \pm\sqrt{3}$$

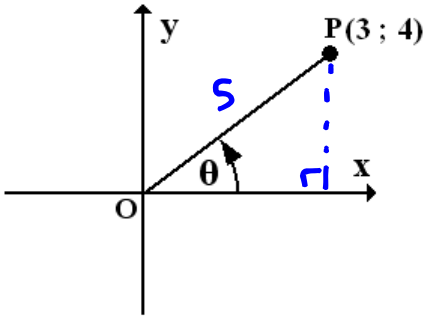
Step 4: Answer the questions:

$$(i)\ \sin\theta = \frac{\pm\sqrt{3}}{2}$$

$$(ii)\ \tan^2\theta = \left(\frac{\pm\sqrt{3}}{1}\right)^2 = 3$$

EXERCISE 10 "Pythag Sketches" in Quadrant 1

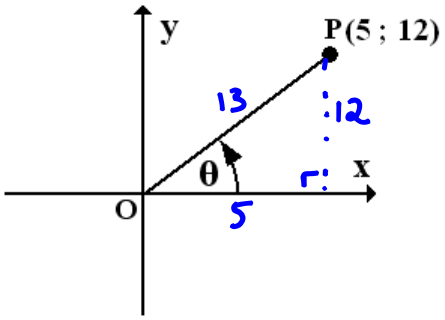
1. Refer to the diagram alongside. OP has been rotated through θ degrees such that P is the point (3 ; 4).



   Without the use of a calculator, determine the value of the following:

   a. OP 5 (Pythag)
   b. sin θ 4/5
   c. cos θ 3/5
   d. tan θ $\frac{4}{3}$

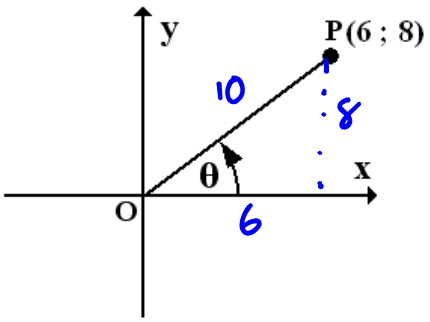
2. Refer to the diagram alongside. OP has been rotated through θ degrees such that P is the point (5 ; 12).



   Without the use of a calculator, determine the value of the following:

   a. OP = 13 (Pythag)
   b. sin θ 12/13
   c. cos θ 5/13
   d. tan θ $\frac{12}{5}$

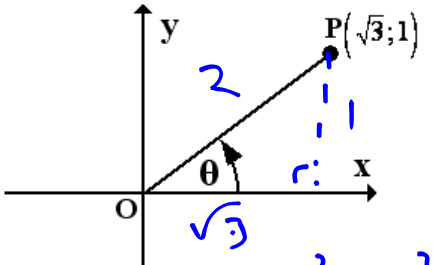
3. Refer to the diagram alongside. OP has been rotated through θ degrees such that P is the point (6 ; 8).



   Without the use of a calculator, determine the value of the following:

   a. OP = 10 (Pythag)
   b. sin θ + cos θ $\frac{8}{10} + \frac{6}{10} = \frac{14}{10} = \frac{7}{5}$
   c. tan θ $= \frac{8}{6} = \frac{4}{3}$

4. Refer to the diagram alongside. OP has been rotated through θ degrees such that P is the point $\left(\sqrt{3};1\right)$.

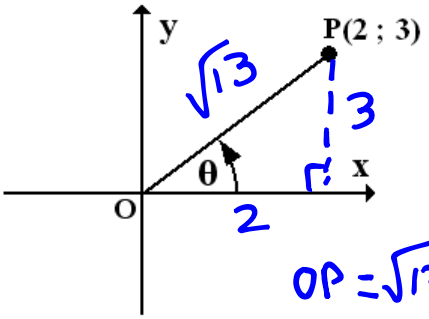


   Without the use of a calculator, determine the value of the following:

   a. sin θ $= \frac{1}{2}$
   b. cos θ $= \frac{\sqrt{3}}{2}$
   c. $\tan^2\theta$ $= \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$

Pythag $x^2 + y^2 = r^2$
$(\sqrt{3})^2 + 1^2 = r^2$
$4 = r^2$
$2 = r$

5. Refer to the diagram alongside. OP has been rotated through θ degrees such that P is the point (2 ; 3).



OP = √13 (Pythag)

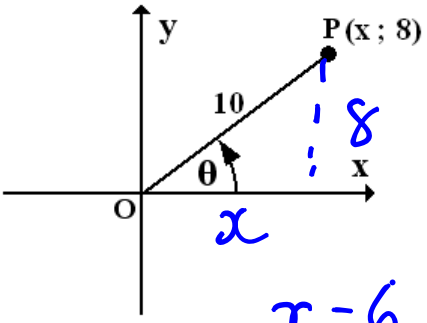
Without the use of a calculator, determine the value of the following:

a. sin θ $= \frac{3}{\sqrt{13}}$

b. cos θ . tan θ

$= \frac{2}{\sqrt{13}} \times \frac{3}{2} = \frac{3}{\sqrt{13}}$

6. Refer to the diagram alongside. OP has been rotated through θ degrees such that OP = 10units and P is the point (x ; 8).



x = 6 (Pythag)

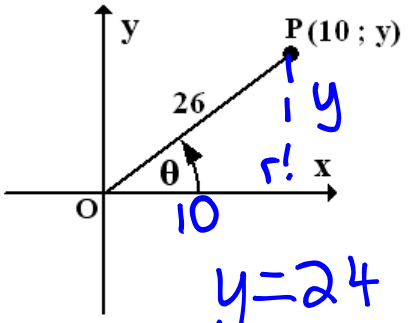
Without the use of a calculator, determine the value of the following:

a. sin θ $= 8/10 = 4/5$

b. $\cos^2\theta$ $(8/10)^2 = \frac{36}{100} = \frac{9}{25}$

c. 12.tan θ $= 12 \times 8/6 = 16$

7. Refer to the diagram alongside. OP has been rotated through θ degrees such that OP = 26units and P is the point (10 ; y).



y = 24 (Pythag)

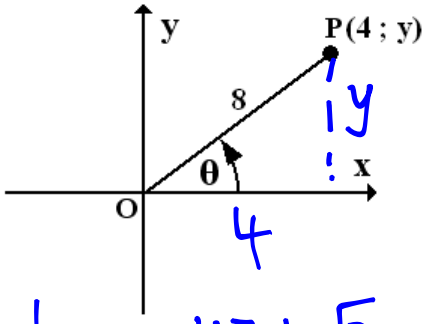
Without the use of a calculator, determine the value of the following:

a. tan θ $= \frac{24}{10} = \frac{12}{5}$

b. $\sin^2\theta + \cos^2\theta$

$= \left(\frac{24}{26}\right)^2 + \left(\frac{10}{26}\right)^2 = 1$

8. Refer to the diagram alongside. OP has been rotated through θ degrees such that OP = 8units and P is the point (4 ; y).



y = 4√3 (Pythag)

Without the use of a calculator, determine the value of the following:

a. tan θ $= \frac{4\sqrt{3}}{4} = \sqrt{3}$

b. $\sin^2\theta + \cos^2\theta$ $= \left(\frac{4\sqrt{3}}{8}\right)^2 + \left(\frac{4}{8}\right)^2 = 1$

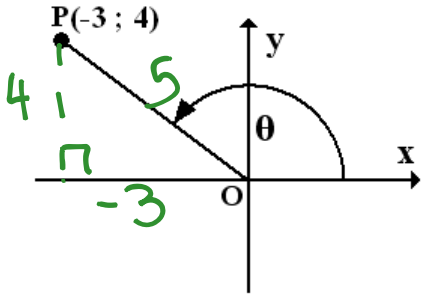
c. $\frac{\sin\theta}{\cos\theta}$

$= \left(\frac{4\sqrt{3}}{8}\right) \div \frac{4}{8} = \frac{\sqrt{3}}{2} \times \frac{8}{4}$

$= \sqrt{3}$

EXERCISE 11 "Pythag sketches" in Quadrants 2, 3 and 4

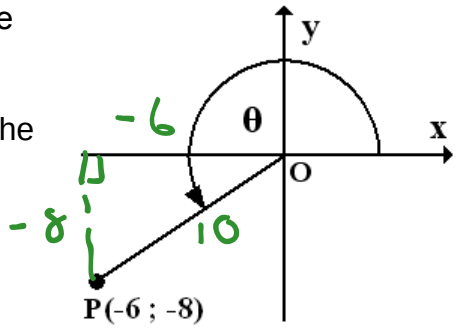
1. Refer to the diagram alongside. OP has been rotated through θ degrees such that P is the point (–3 ; 4).



   Without the use of a calculator, determine the value of the following:

   a. OP = 5 (Pythag)
   b. sin θ = 4/5
   c. cos θ = −3/5
   d. tan θ = 4/−3

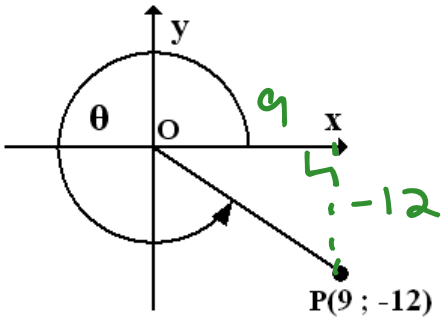
2. Refer to the diagram alongside. OP has been rotated through θ degrees such that P is the point (–6 ; –8).



   Without the use of a calculator, determine the value of the following:

   a. tan θ = −8/−6 = 4/3
   b. OP = 10 (Pythag)
   c. sin θ = −8/10 = −4/5
   d. 100.cos θ = 100(−6/10) = −60

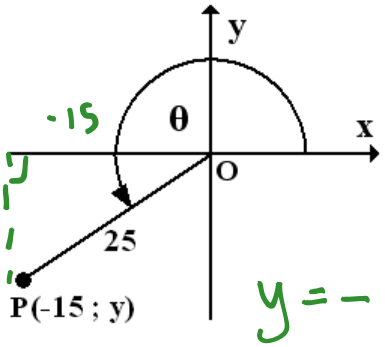
3. Refer to the diagram alongside. OP has been rotated through θ degrees such that P is the point (9 ; –12).



   Without the use of a calculator, determine the value of the following:

   a. 18.tan θ = 18(−12/9) = −24
   b. OP = 15 (Pythag)
   c. sin θ + cos θ = −12/15 + 9/15 = −3/15 = −1/5

4. Refer to the diagram alongside. OP has been rotated through θ degrees such that OP = 25 units and P is the point (–15 ; y).

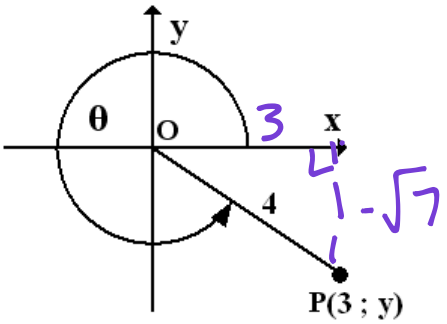


   Without the use of a calculator, determine the value of the following:

   a. 50.sin θ = 50(−20/25) = −40
   b. cos θ = −15/25 = −3/5
   c. $\tan^2\theta$ = (−20/−15)² = 16/9

y = −20 (Pythag)

5. Refer to the diagram alongside. OP has been rotated through θ degrees such that OP = 4 units and P is the point (3 ; y).

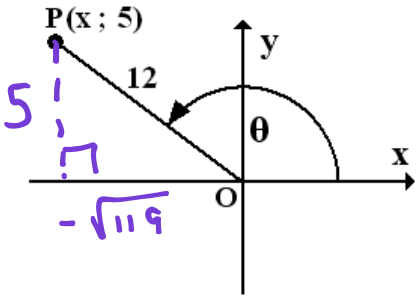


Without the use of a calculator, determine the value of the following:

a. sin θ

b. cos θ . tan θ

6. Refer to the diagram alongside. OP has been rotated through θ degrees such that OP = 12 units and P is the point (x ; 5).



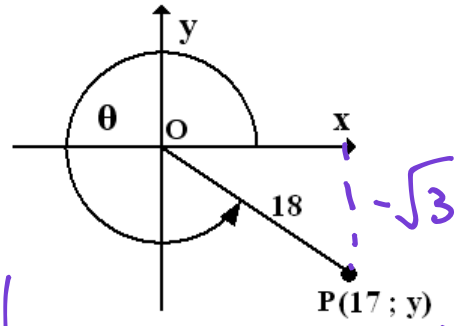
Without the use of a calculator, determine the value of the following:

a. sin θ

b. $\cos^2\theta$

c. tan θ

7. Refer to the diagram alongside. OP has been rotated through θ degrees such that OP = 18 units and P is the point (17 ; y).

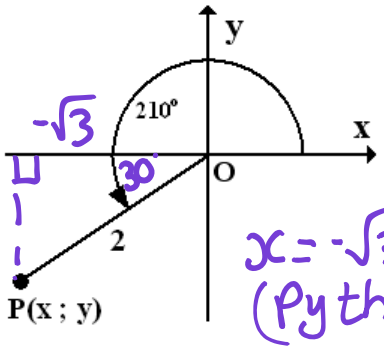


Without the use of a calculator, determine the value of the following:

a. cos θ

b. $\sin^2\theta + \cos^2\theta$

8. Refer to the diagram alongside. OP has been rotated through 210º such that OP = 2 units and P is the point (x ; y).



Without the use of a calculator, determine the value of the following:

a. –6.sin θ

b. cos θ

c. tan θ

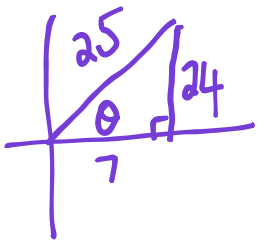
EXERCISE 12

For each of the following questions:
Draw a rough sketch to represent the given information on a Cartesian plane.
Fill in the *x*, *y* and *r* values on your sketch and indicate the angle.
Determine the value of the questions that follow (without the use of a calculator).

* Give Pythag as a reason each time *

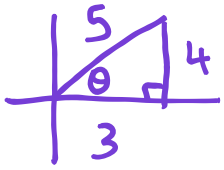
1. If $\cos\theta = \frac{7}{25}$ and $\theta \in [0^\circ;90^\circ]$ determine:

   a) $\tan\theta$ b) $\sin\theta$ c) $\tan\theta.\cos\theta$



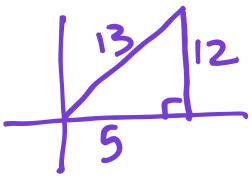
2. If $\sin\theta = \frac{4}{5}$ and $\theta \in [0^\circ;90^\circ]$ determine:

   a) $\cos^2\theta$ b) $\sin\theta + \cos\theta$ c) $3\tan\theta$



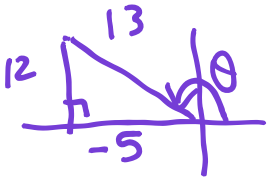
3. If $5\tan\theta = 12$ and $0^\circ \le \theta \le 90^\circ$, determine:

   a) $\sin\theta$ b) $(\sin\theta + \cos\theta)^2$ c) $26\cos\theta - 3$



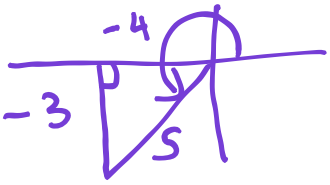
4. If $\sin\hat{A} = \frac{12}{13}$ and $\hat{A} \in [90^\circ;360^\circ]$ determine:

   a) $\cos\hat{A}$ b) $1 - \sin^2\hat{A}$



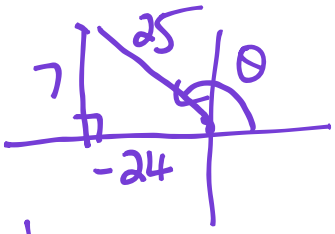
5. If $\tan\hat{P} = \frac{3}{4}$ and $180^\circ \le \hat{P} \le 360^\circ$ determine:

   a) $\sin\hat{P}.\cos\hat{P}$ b) $\frac{-4\tan\hat{P}}{\sin\hat{P}}$



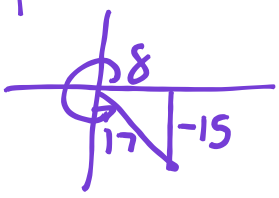
6. If $\cos\alpha = -\frac{24}{25}$ and $\sin\alpha > 0$ determine:

   a) $-5\cos\alpha.\tan\alpha$ b) $\frac{1}{\sin^2\alpha} - \frac{1}{\tan^2\alpha}$



7. If $8\tan\beta + 15 = 0$ and $\cos\beta > 0$ determine:

   a) $17\sin\beta + 6$ b) $\frac{15}{\tan\beta}$



8. If $\sin\theta = -\frac{6}{7}$ and $\theta \in [90^\circ;270^\circ]$ determine:

   a) $\cos\theta$ b) $\tan^2\theta$

